

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati | github.com/deepak-glitch |
Kent, OH (US-based, F-1 OPT)

PROFESSIONAL SUMMARY

Applied AI engineer with 2.5+ years shipping production agentic LLM systems in regulated healthcare - tool-calling agents, RAG pipelines, HIPAA-conscious data governance, schema-validated multi-agent contracts, audit trails. Masters degree (Kent State). Founding-team mindset: shipped Manga Lens to Chrome Web Store solo with four AI vision providers behind a single abstraction; ran a 5-stage Healthcare Claims pipeline with JSON-validated handoffs to prevent cascading hallucinations; founded an AgriTech marketplace and onboarded 80-120 active users in phase one. Comfortable owning end-to-end from prompt to production.

CORE COMPETENCIES

Agentic LLM workflows • Tool calling • RAG • Embeddings • Vector search • Multi-provider abstraction (Claude + OpenAI) • HIPAA-conscious data pipelines • Audit trails • FastAPI/Flask • Docker • TypeScript • Postgres/SQL • Python • Event-driven backend • Schema-validated agent contracts • Production observability • 0->1 ownership

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee

USA - Healthcare Technology, AI Consulting, SaaS Healthcare

- Developed **agentic LLM workflows** for multi-step healthcare queries (eligibility checks, care navigation, documentation clarification) with structured reasoning, tool discipline, and grounding rules; improved agent response stability by **~25%**.
- Built **Retrieval-Augmented Generation (RAG) systems** for clinical knowledge retrieval; improved retrieval precision **~35%**, reduced irrelevant retrieval **>30%**, and pushed retrieval-grounded response alignment above **90%**.
- Packaged ML/LLM inference as **FastAPI/Flask** RESTful services, containerized with **Docker**, with structured logging and load simulation; reduced post-deployment defects by **~30%**.
- Shipped **predictive ML pipelines** (scikit-learn, XGBoost) for patient no-show prediction and care engagement scoring; improved recall **15-20%** on high-risk categories while holding precision **>90%**.
- Built preprocessing pipelines for EHR extracts and support tickets; raised dataset reliability above **98%** and cut downstream model instability by **~40%**.
- Maintained HIPAA-conscious data governance: de-identification, data lineage docs, evaluation audit trails, and stakeholder-facing system-limitation docs.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern)

Hyderabad, India - Oil & Gas, Enterprise ERP, Industrial Automation

- Optimized SQL and **T-SQL stored procedures** powering the Synthesis Order-to-Cash platform; reduced query execution time **~20%** and data retrieval latency **~25%**.
- Built **CI/CD pipelines with Jenkins** for schema updates and stored procedure deployments; reduced deployment errors **>30%** and improved release cycle time **~35-40%** via structured release validation and rollback checkpoints.
- Designed performance dashboards using SQL Server DMVs and Grafana to track long-running queries, deadlocks, CPU/I/O contention; cut incident recurrence **~25%** and improved root-cause resolution speed **~30%**.
- Supported RBAC implementation and audit logging for financial modules in a compliance-sensitive oil & gas environment.

- Built a **transformer-based video summarization and highlight-generation system** across 5,000+ recorded sessions; reduced manual review time by 60-70%.
- Implemented **document Q&A with RAG** (semantic chunking + embedding retrieval); reduced hallucinated outputs ~30% and raised contextual answer accuracy >85%.
- Deployed the stack as a **Flask-based API** with a lightweight web interface for non-technical staff.

PROJECTS

Agentic Healthcare Claims Processing & Fraud Risk Intelligence System

Multi-agent pipeline (Intake Normalization to Validation to Consistency to Duplicate Detection to Fraud Risk Scoring) with schema-validated JSON contracts between agents to prevent cascading hallucinations. RAG-grounded CPT/ICD validation against vector-indexed policy documents; ANN similarity search for duplicate claim detection; explainable risk scoring with audit-ready reasoning traces.

Manga Lens - Chrome Extension for Real-Time AI Manga/Webtoon Translation shipped, Chrome Web Store

Full-stack browser extension built solo (Manifest V3, content scripts, service workers, captureVisibleTab). Integrates four AI vision providers (Claude Sonnet, GPT-4o mini, GPT-4.1 Nano, Ollama/minicpm-v local). Multi-section capture pipeline for tall webtoon panels with viewport-slice screenshots, coordinate remapping, and dedup. Seven-day translation cache, per-domain selector configs for 29 sites, narrowed host permissions, multi-provider payload handling (WebP cloud, JPEG Ollama).

Dream Decoder - AI-Powered Multimodal Creative Intelligence Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind app that interprets dreams, generates poetic reinterpretations, and synthesizes dreamscapes via coordinated multimodal APIs (Perplexity Sonar for interpretation, Replicate image models for 16:9 visuals). Intermediate structured prompt transformation layers raised contextual alignment ~30% and first-pass image success rate ~25-30% over naive direct-prompt orchestration.

Agentic Pixel Character Synthesis & Animation Engine ongoing

Phased generative AI system for identity-consistent, pose-controlled pixel character synthesis. Stable Diffusion fine-tuning + LoRA for identity, ControlNet + OpenPose/MediaPipe for pose, latent-space consistency for animation, LLM-based agent orchestrator that decomposes high-level prompts into generation tasks. Backend: PyTorch, Hugging Face Diffusers, FastAPI, Docker.

E-Farming Digital Marketplace (Founder & Full-Stack Developer)

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace connecting small-scale farmers with buyers. Onboarded 80-120 active users in phase one; cart, reviews, and community blogging features to drive repeat engagement.

Patient Records Management Web Application

Bottle + SQLite relational healthcare CRUD app with normalized patients/diagnoses schema, FK constraints, application-layer cascade deletes, and LEFT JOIN + GROUP_CONCAT aggregation for single-query patient+diagnosis retrieval.

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

CERTIFICATIONS

F-1 OPT - authorized to work in the United States; future sponsorship required

SKILLS

Languages & Frameworks: Python, FastAPI, Flask, SQL (T-SQL, PostgreSQL, SQLite), TypeScript, React, C++ (Arduino), PHP

LLM & GenAI: LLM application development, RAG, agentic workflows, structured outputs, prompt engineering, evaluation pipelines, guardrails, grounding, semantic retrieval, embeddings, vector search, responsible AI

ML Libraries: scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, Diffusers, LangChain, LlamaIndex

Computer Vision & Multimodal: YOLOv8, OpenCV, ControlNet, OpenPose/MediaPipe, Stable Diffusion, video analytics

Data & Analytics: Pandas, NumPy, feature engineering, time-series forecasting, walk-forward validation, NetworkX

Infra & DevOps: Docker, Jenkins, Grafana, CI/CD, MLOps/LLMOps, RESTful APIs, cloud-ready deployment, observability/logging

Domains: Healthcare AI (HIPAA-conscious), Applied AI, Enterprise AI, Workflow Automation, Computer Vision, Multimodal Systems